

# Omin Kwon

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## RESEARCH INTEREST

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Computer Architectures & Machine Learning Systems, with an emphasis on co-designing ML algorithms and computer systems for efficient ML.

## EDUCATION

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### Seoul National University

Sept 2025 – present

*M.S. in Computer Science and Engineering*

- Advisor: Prof. Jae W. Lee, ARC Lab
- GPA: 4.3/4.3

### Seoul National University

Mar 2019 – Aug 2025

*B.S. in Electrical and Computer Engineering (summa cum laude)*

- GPA: 4.26/4.3 (upper-division), 3.95/4.3 (overall)
- Leave of absence for mandatory military service (May 2021-Feb 2023)

## PUBLICATIONS

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### HERALD: High Throughput Block Diffusion LLM Serving via CPU-GPU Cooperative KV Cache Retrieval

Omin Kwon, Doyeon Kim, Jongseok Park, Seung Yul Lee, Ion Stoica, Jae W. Lee

Under review

Paper Link: [arxiv.org/abs/2606.21633](https://arxiv.org/abs/2606.21633)

### ELIXIR: Disaggregating FFN, Attention, and KV Cache via CXL-PNM for Elastic LLM Serving

Omin Kwon\*, Kyungsoo Kim\*, Ugyeong Song, Yeonhong Park, Nachuan Wang, Namsung Kim, Jae W. Lee

Under review

### MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM

Omin Kwon, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee

ICML 2026 AdaptFM Workshop (Oral); extended version under review

Paper Link: [arxiv.org/abs/2602.14209](https://arxiv.org/abs/2602.14209)

### STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer

Omin Kwon\*, Kyungjun Oh\*, Jaeyong Lee, Myungsuk Kim, Jihong Kim

ICCAD 2025

Paper Link: [doi.org/10.1109/ICCAD66269.2025.11240772](https://doi.org/10.1109/ICCAD66269.2025.11240772)

### NestedFP: High-Performance, Memory-Efficient Dual-Precision Floating Point Support for LLMs

Haeun Lee, Omin Kwon, Yeonhong Park, Jae W. Lee

NeurIPS 2025

Paper Link: [neurips.cc/virtual/2025/loc/san-diego/poster/117613](https://neurips.cc/virtual/2025/loc/san-diego/poster/117613)

### AiDE: Attention-FFN Disaggregated Execution for Cost-Effective LLM Decoding on CXL-PNM

KyungSoo Kim, Omin Kwon, Yeonhong Park, Jae W. Lee

IEEE Computer Architecture Letters (CAL)

Paper Link: [doi.org/10.1109/LCA.2025.3597323](https://doi.org/10.1109/LCA.2025.3597323)

## EXPERIENCES

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### Visiting Researcher

Mar 2026 – May 2026

University of Illinois Urbana-Champaign (UIUC)

Urbana, IL, USA

- FAST Lab, hosted by Prof. Nam Sung Kim

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|--------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Microelectronics (430.207B) Tutor</b><br>Seoul National University                | Mar 2025 – June 2025<br>Seoul, South Korea |
| <b>Digital System Design (430.315) Tutor</b><br>Seoul National University            | Sept 2024 – Dec 2024<br>Seoul, South Korea |
| <b>Computer Organization and Design (430.322) Tutor</b><br>Seoul National University | Mar 2024 – June 2024<br>Seoul, South Korea |
| <b>Military Service</b><br>Republic of Korea Air Force                               | May 2021 – Feb 2023<br>South Korea         |

**PRESENTATIONS**

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| <b>MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM</b><br>Oral paper at the ICML 2026 AdaptFM Workshop (1.5%). arXiv.                    | 2026 |
| <b>STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer</b><br>Oral presentation at ICCAD 2025, Munich. Paper. | 2025 |

**HONORS & AWARDS**

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| <b>Grand Prize (Minister of Science and ICT Award), 2024 AI Semiconductor Idea and Design Contest</b><br>Ministry of Science and ICT<br>Designed a resolution-aware diffusion accelerator leveraging computational staleness. News: Edaily.                                    | Jan 2025            |
| <b>Grand Prize (Deputy Prime Minister and Minister of Education Award), 2024 STOB League Semiconductor Contest</b><br>Ministry of Education<br>Designed a state-aware data randomizer for high-performance SSDs, improving performance and device lifetime. News: Dong-A Ilbo. | Dec 2024            |
| <b>Grand Prize, 2024 FriendliAI LLM Hackathon</b><br>FriendliAI<br>Developed a knowledge-graph RAG application using the FriendliAI API.                                                                                                                                       | July 2024           |
| <b>SNU Specialized Semiconductor College Scholarship</b><br>Seoul National University                                                                                                                                                                                          | Aug 2024 – Aug 2025 |
| <b>Merit-Based Scholarship</b><br>Seoul National University                                                                                                                                                                                                                    | 2023, 2024          |